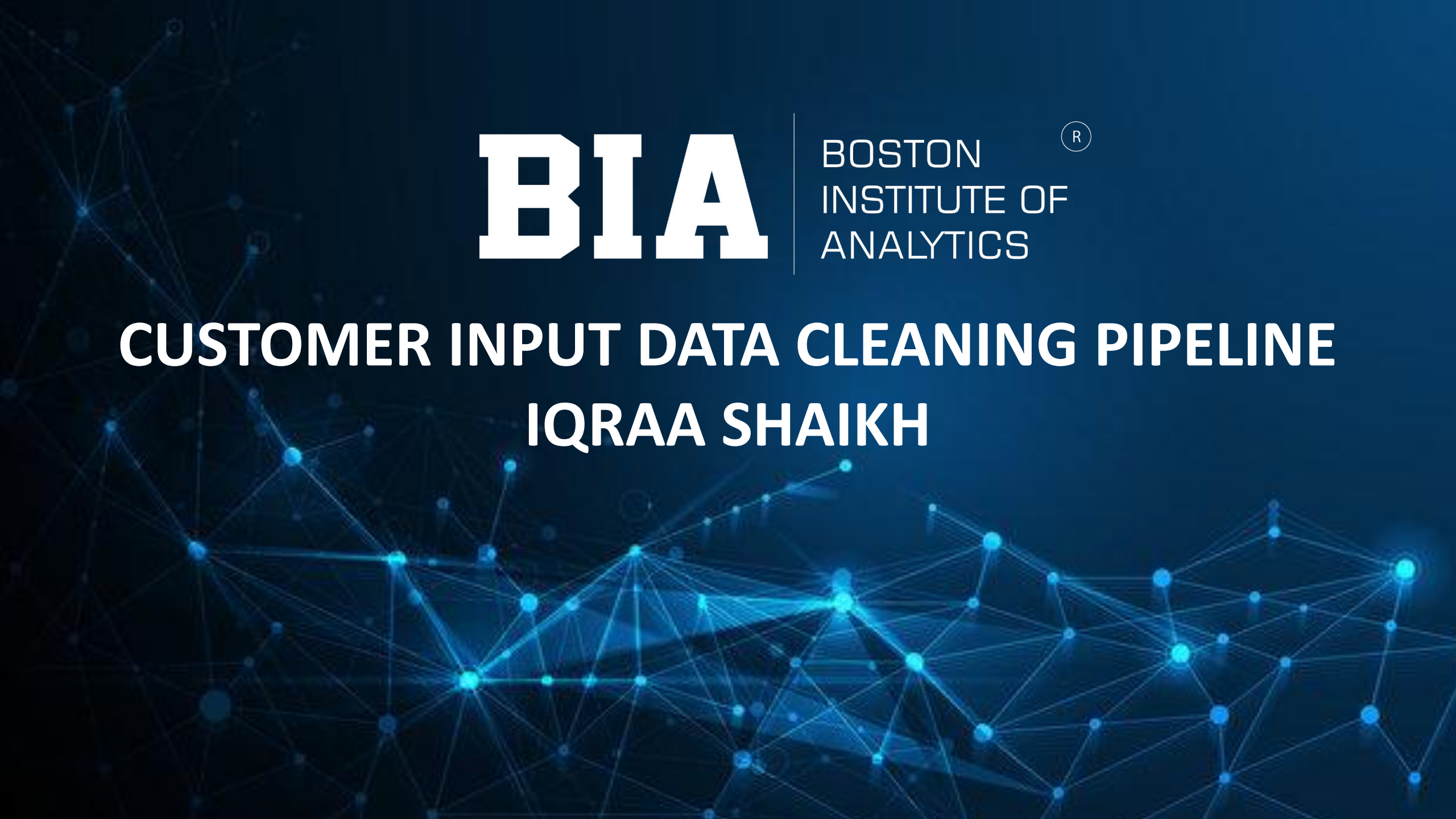




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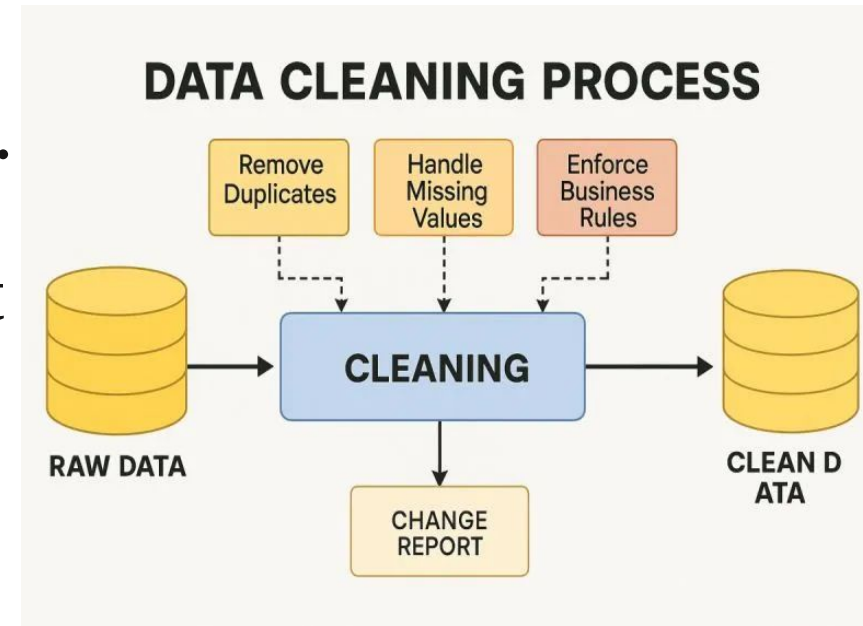
CUSTOMER INPUT DATA CLEANING PIPELINE

IQRAA SHAIKH

AGENDA:

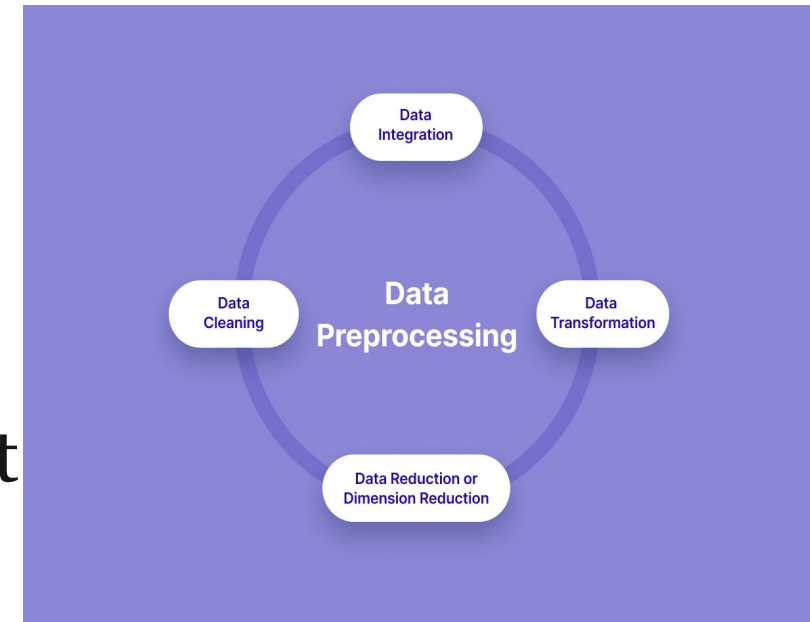
This project aims to convert raw customer comments into a clean, analysis-ready format. Users provide an Excel and CSV file containing a Comment column, and the script outputs a cleaned version of the comments suitable for downstream tasks such as sentiment analysis, topic modeling, or keyword extraction.

The pipeline focuses on **rule-based text processing**, not machine learning.



INTRODUCTION:

In many customer-facing applications, users provide open-text feedback that contains valuable insights about their experiences, expectations, and concerns. However, these raw comments often contain noise such as emojis, HTML tags, repeated characters, and inconsistent formatting, making them unsuitable for direct analysis. Before meaningful insights can be extracted—whether through sentiment analysis, keyword extraction, or automated categorization. The text must be cleaned and standardized.



OBJECTIVES:

The primary objective of this project is to develop a **fully automated text cleaning pipeline** that prepares customer comments for downstream analytics. Specific goals include:

1. **Build a robust cleaning system** capable of transforming raw textual data into clean, consistent, and analysis-ready format
2. **Implement modular cleaning steps** such as emoji removal, HTML tag stripping, contraction expansion, and removal of non-alphabetical characters.
3. **Normalize and standardize text** through repeated character reduction, lowercasing, and stopword removal.
4. **Ensure scalability**, enabling the pipeline to efficiently process 1000+ comments columns for easy comparison.
5. **Produce high-quality output files** (excel/CSV) with original and cleaned comments in under 2-minutes.
6. **Add optional enhancement** such as success flags, word count comparisons, and a simple GUI interface.

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SCOPE:

- . Input handling for excel or CSV files
- . Cleaning operations:
 - .Emoji removal
 - .Expanding contractions
 - .Normalizing repeated characters
 - .Removing non-alphabetic characters
 - .Lowercasing
 - .Tokenization and stopword removal
- . Creation of a cleaned output file(.csv/xlsx)
- . Logging and basic validation
- . Optional metadata columns (word count before/after, success flag)

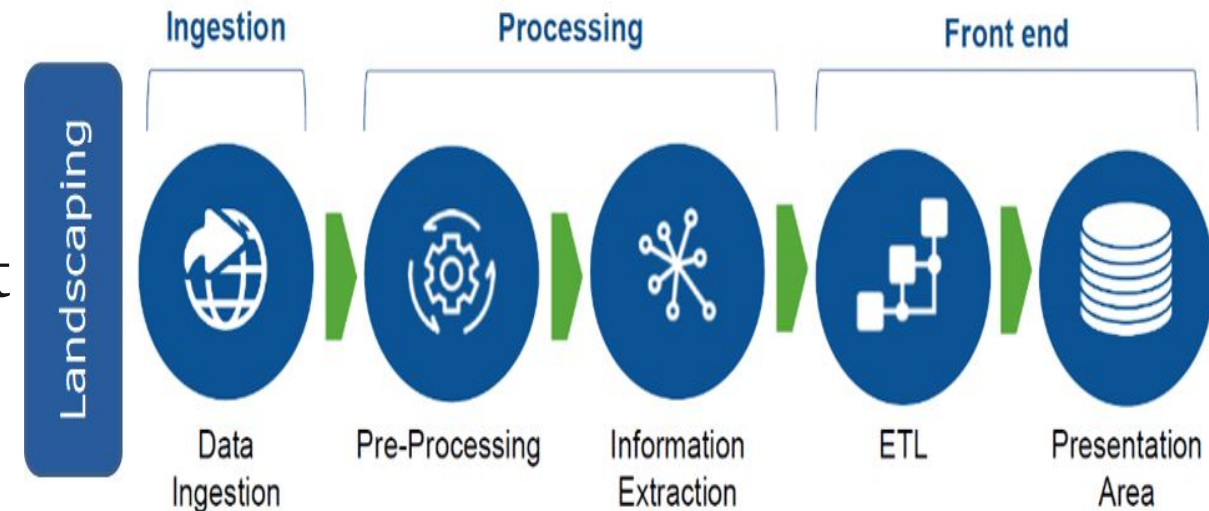
Data Cleaning and Preprocessing Techniques



OUT OF SCOPE:

- . Machine Learning model development
- . Sentiment analysis or topic modeling
(pipeline prepares for these tasks but does not include them)
- . Multi-language support
(English only unless extended)
- . Database storage or cloud deployment

Data pipeline

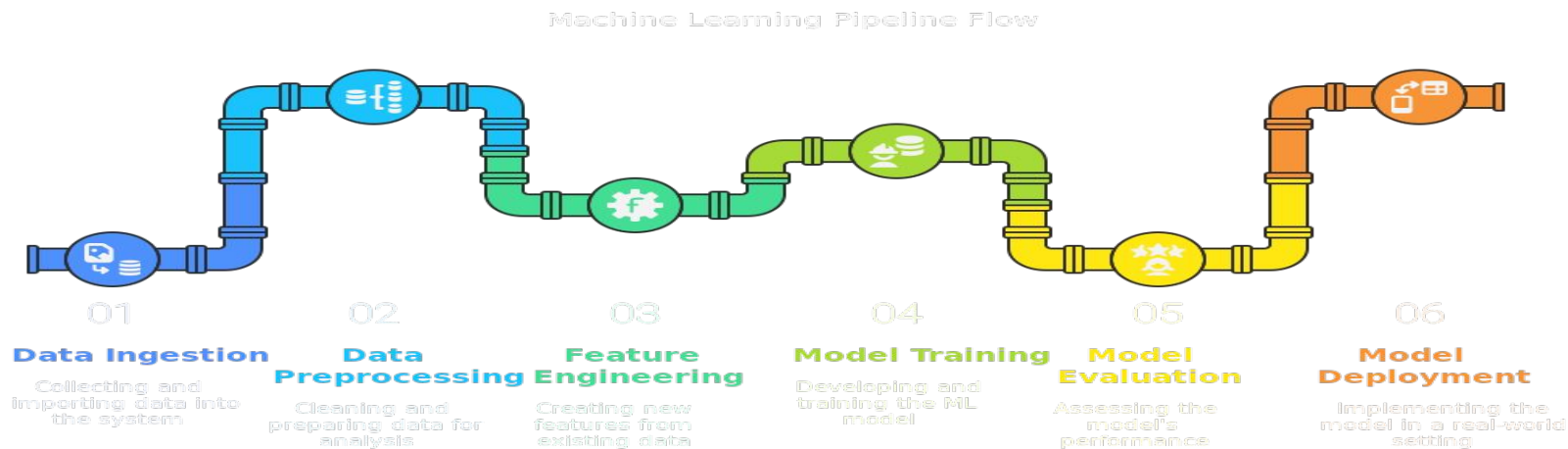


METHODOLOGY:

The project follows a structured and modular methodology to ensure the efficient transformation of raw customer comments into analysis-ready text. The process is divided into the following steps:

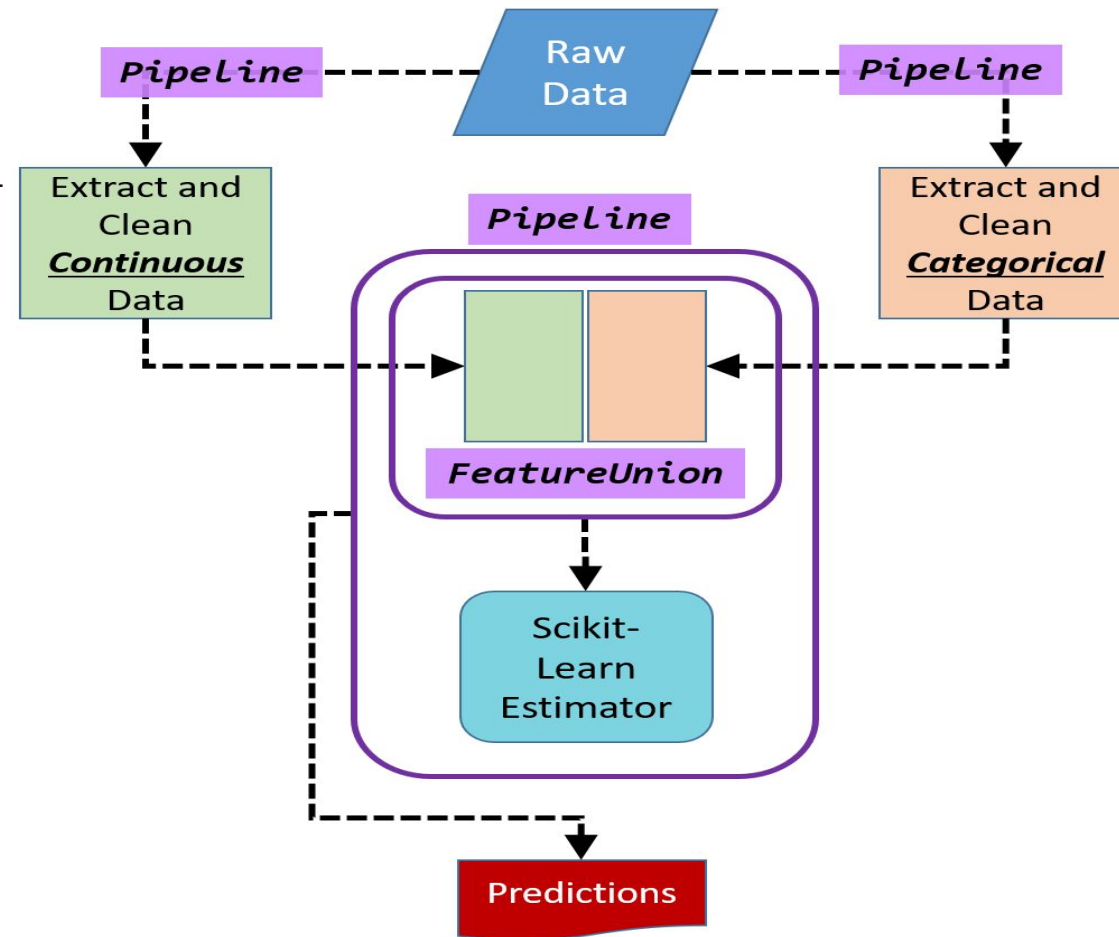
1. REQUIREMENT ANALYSIS:

- . Identify the expected input and desired output
- . Define cleaning rules based on common text issues (emojis, HTML, contractions, etc)
- . Confirm processing constraints such as runtime efficiency



2. DATA INGESTION:

- . Load Excel/CSV file using pandas
- . Validate the presence of the required **Comment** column
- . Perform basic data checks
(null values, duplicates)



3. TEXT CLEANING PIPELINE:

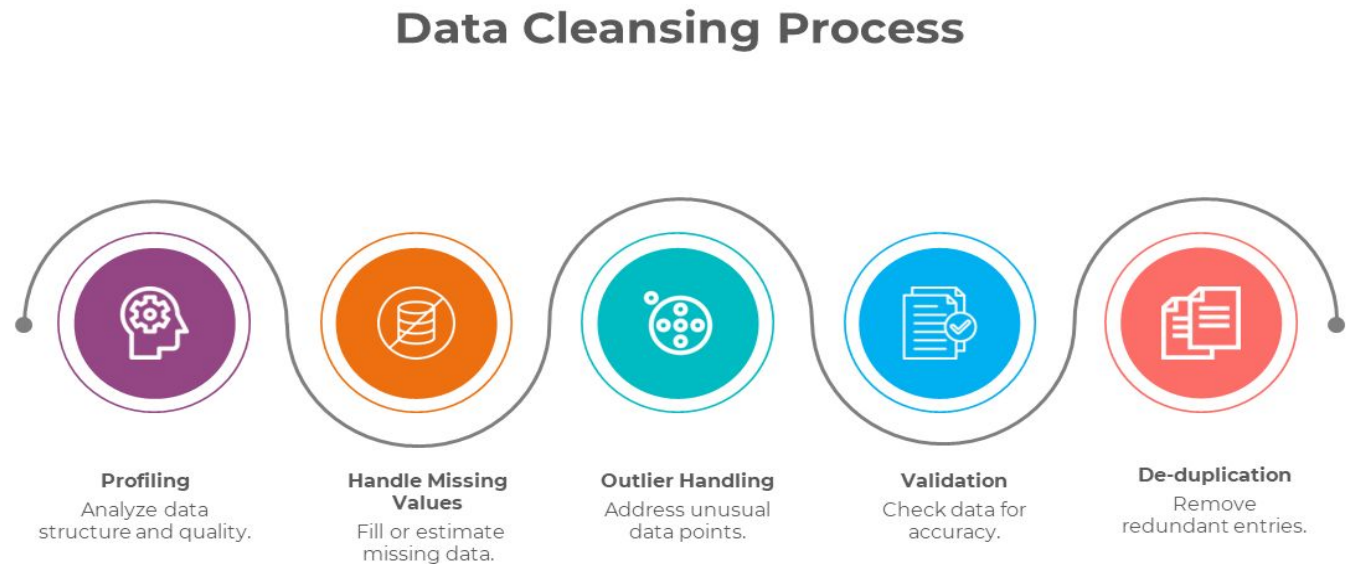


A modular sequence of function performs specific cleaning steps:

- . Emoji Removal: Remove all unicode emoji characters.
- . HTML Tag Stripping: Use BeautifulSoup to extract text from HTML.
- . Contraction Expansion: Convert shortened forms to standard English.
- . Character Normalization: Reduce repeated character to normal length.
- . Symbol Cleaning: Remove all non-alphabetic character using regex.
- . Lowercasing: Normalize letter casing across entries.
- . Tokenization: Split text into tokens for controlled processing.
- . Stopword Removal: Filter out non-informative words.

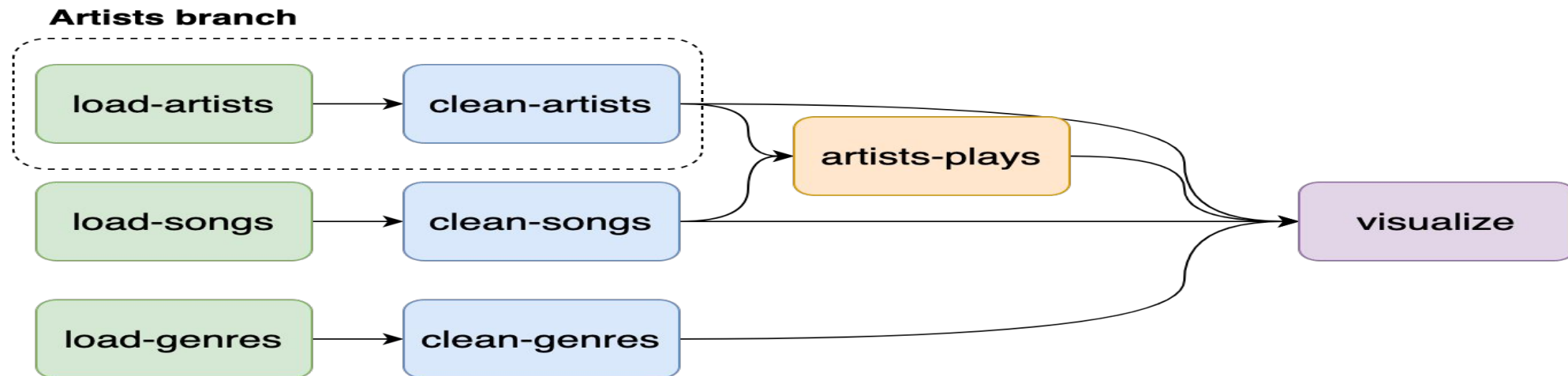
4. OUTPUT ENHANCEMENT:

- . Add a Cleaned_Comment column
- . Add optional columns:
 - .Cleaning_Success
 - .WordCount_Before
 - .WordCount_After



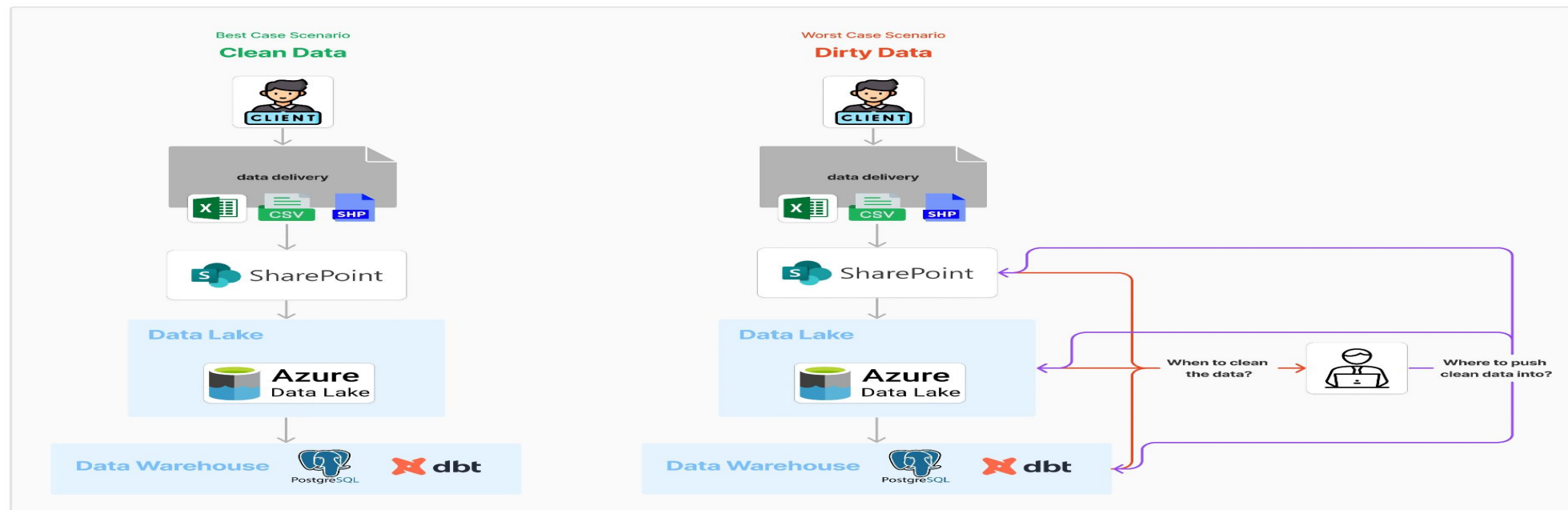
5. OUTPUT GENERATION:

- .Export the cleaned dataset to Excel or CSV.
- . Ensure file formatting and naming consistency.



6.PERFORMANCE TESTING:

- . Validate speed using dataset with 1000+ comments.
- . Ensure consistent functionally across different file types.



SYSTEM ARCHITECTURE:

The system architecture is designed to be simple, modular, and extensible. It includes the following components:

1. Input Layer

- . User provides and Excel/CSV file containing raw comments

2. Data Processing Layer

- . File loader (pandas)
- . Sequential cleaning functions
- . Logging and error handling

3. NLP Layer

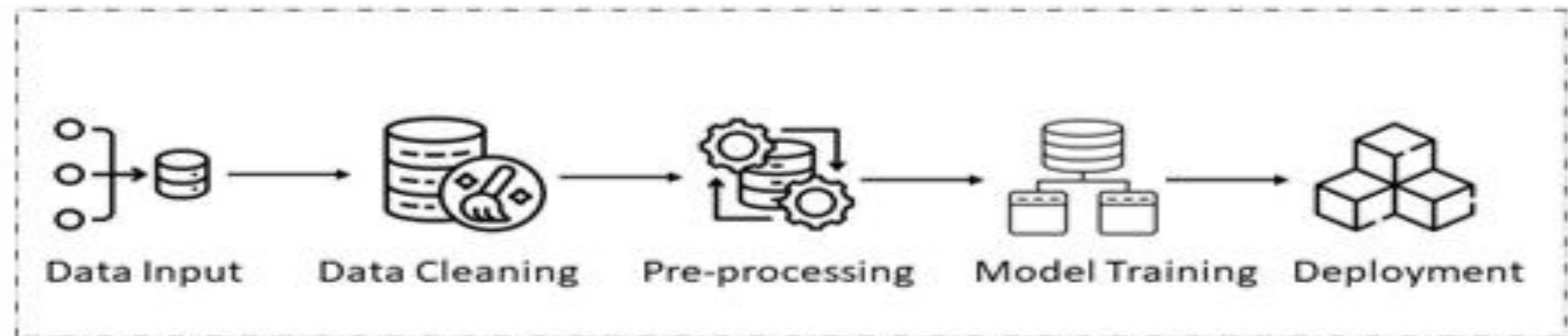
- . Tokenization
- . Stopword filtering
- . Contraction expansion

4. Output Layer

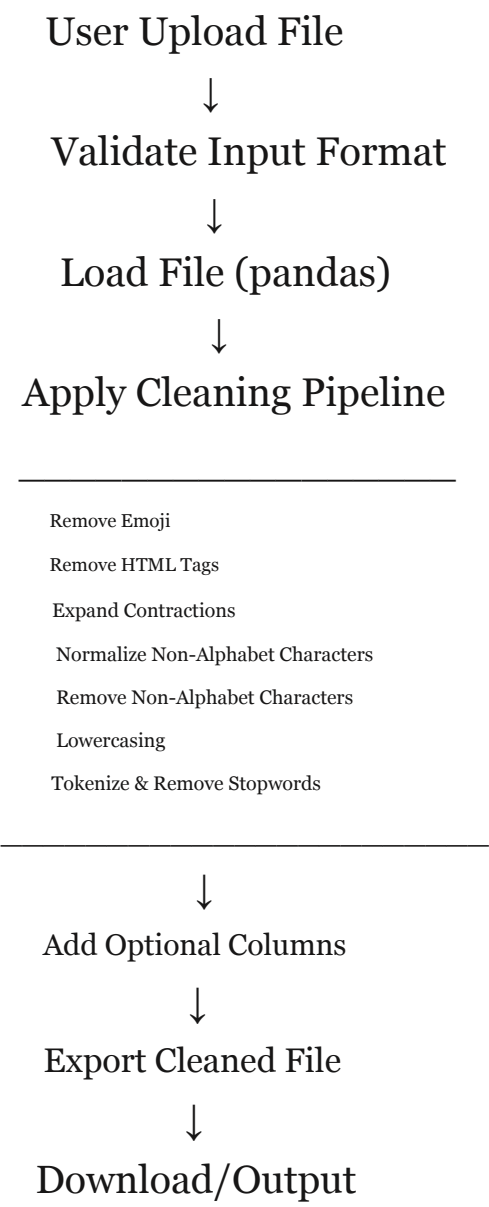
- . Cleaned file export
- . Optional summary metrics
- . GUI interface (optional)

5. Interaction Layer (Optional GUI)

- . File upload and preview
- . Run cleaning pipeline
- . Download cleaned file



FLOWCHART:



TOOLS & TECHNOLOGIES:

Programming Language

.Python 3.x

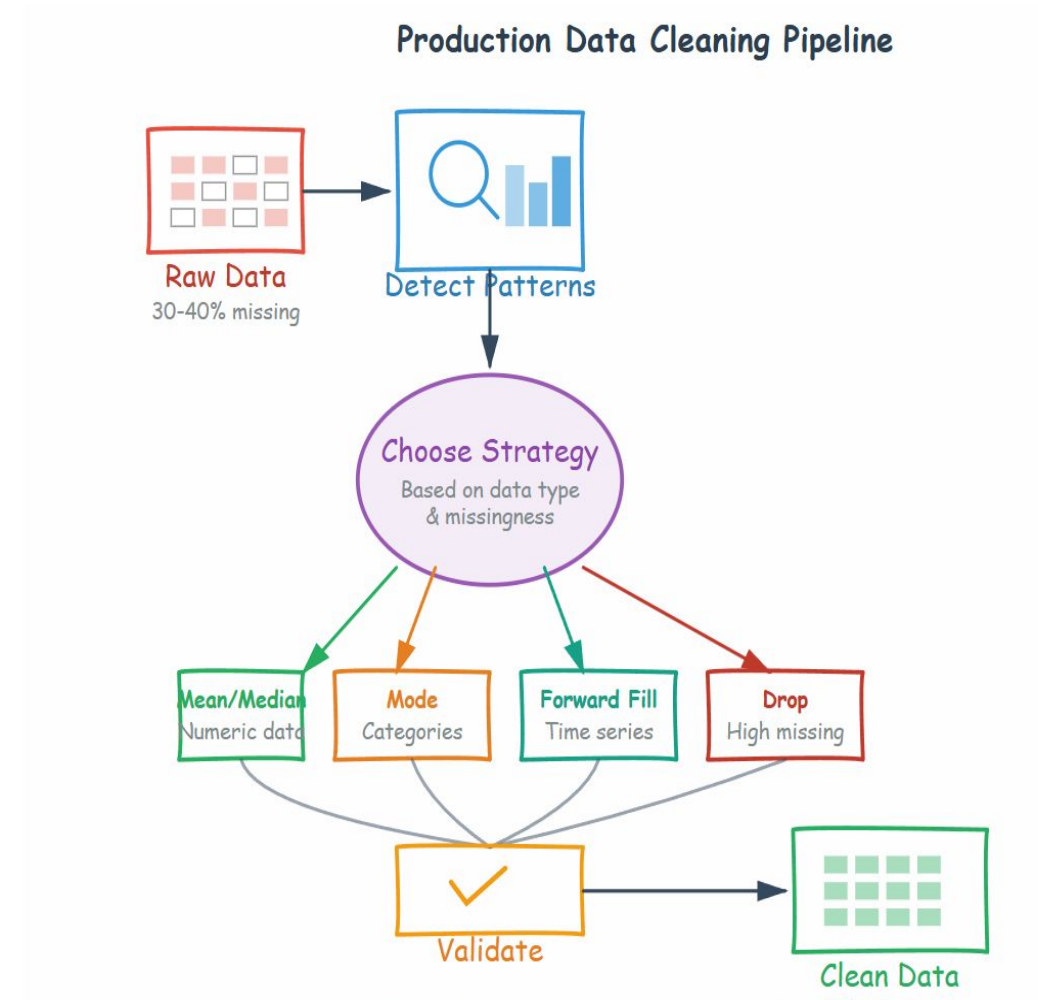
Libraries

- .Pandas- File loading and DataFrame manipulation
- .Regex (re)- Pattern matching and text cleanup
- .Emoji- Emoji detection and removal
- .BeautifulSoup(bs4)- HTML tag stripping
- .Contractions- Contraction expansion

File Formats

.Input: xlsx, .csv

.Output: xlsx, .csv



TOOLS & TECHNOLOGIES:

Programming Language

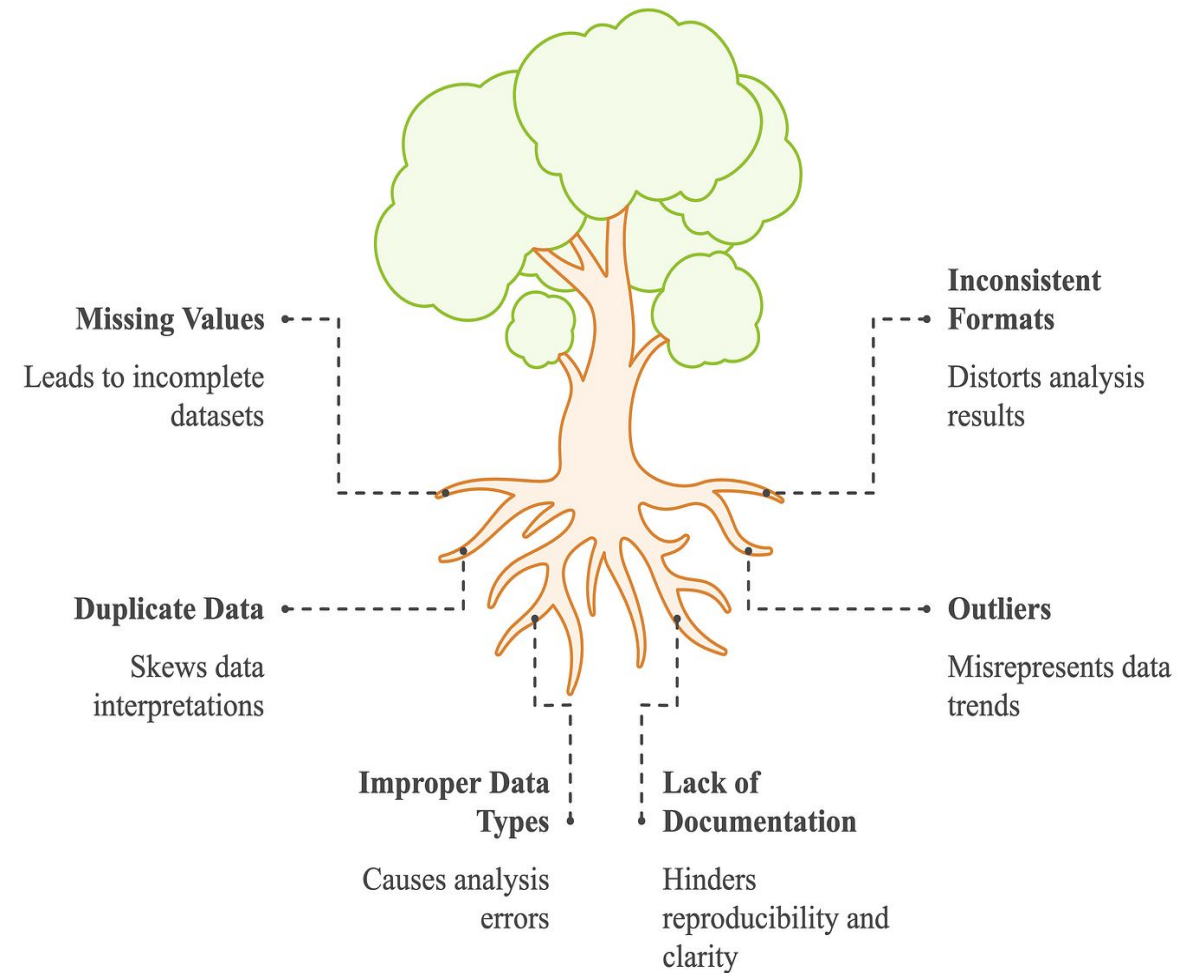
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File Formats

- .Input: xlsx, .csv
- .Output: xlsx, .csv



ENVIRONMENT:

- . Can run locally on any OS with python installed
- . Lightweight and efficient- no GPU or heavy computation required



OBSERVATION:

During the development and execution of the Customer Input Data Cleaning Pipeline, several important observations were made regarding the nature of raw customer comments, the cleaning process, and system performance:

1. Raw Customer Text Contains Significant Noise
2. Modular Cleaning Produces Better Consistency
3. Contraction Expansion and Stopword Removal Improve Clarity
4. Normalizing Repeated Characters Helps Standardization
5. Performance Meets Efficiency Requirements
6. Cleaned Output Is Much More Analysis-Ready
7. Optional Enhancements Add Value



CONCLUSION:

The Customer Input Data Cleaning Pipeline project successfully delivers a reliable and efficient solution for preparing raw customer comments for further analysis. By implementing a modular and rule-based text cleaning system, the pipeline addresses common issues found in user-generated text, such as emojis, HTML, tags, contractions, repeated characters and noise from non-alphabetic symbols. The result is a standardized and analysis-ready dataset that significantly enhances the accuracy and quality of downstream tasks like sentiments analysis, keyword extraction, and topic modeling.

The background is a dark blue gradient with a complex network of glowing blue lines and dots, resembling a molecular structure or a data network. The dots are of varying sizes and brightness, with some appearing as bright blue spheres. The lines connect these dots in a web-like pattern, creating a sense of depth and connectivity.

Thank You!